

AIBS - Research Proposal

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1. Target audience

Our handbook will target small and medium-sized home-care organizations, using Thuiszorg Totaal in Arnhem as our partner-company case which employs over 70 professional healthcare workers. These SMEs have to coordinate many caregivers who visit clients at different locations throughout the day. They face a complicated planning problem because schedules must consider employee availability, client locations, travel time, required care skills, working hours, client preferences, and continuity of care. The specific decision we want to help them make is: Can AI-assisted scheduling help home-care SMEs create more efficient staff schedules, reduce administrative workload and travel time, while maintaining the quality and continuity of care?

2. What they need:

The SME needs to get comfortable with using AI-tools, know what the costs are, what the AI-tool can help with and to train employees to gain an understanding of this new AI-tool/system within the workspace. It is important to keep the human always in the loop. Even a simple AI-supported facilitation of an SME can significantly improve worker efficiency. Thus, improving productivity.

3. Research question.

Main question: Should a small home-care provider like Thuiszorg Totaal adopt an AI supported tool to give clients advance notice of caregiver visits, and what would adopting one actually require?

External analysis

1. What AI-supported scheduling or client-notification tools exist for home care or comparable field-service work and how mature and affordable are they for a Dutch not-for-profit care organisation?

2. What geopolitical and regulatory dynamics, such as US/EU competition over AI infrastructure, data sovereignty, the EU's AI Act and GDPR regime, currently shape which AI vendors a Dutch SME can safely rely on?

2.1 Specifically for a Dutch home-care SME like Thuiszorg Totaal, does it matter whether a scheduling/notification tool is built by a US hyperscaler, a European vendor, or a Dutch care-sector supplier and what does GDPR require in each case?

Internal analysis

1. How does Thuiszorg Totaal currently plan visits and communicate them to clients, and at which point does the information stop moving?
2. Does the admin/IT function have the systems and technical capacity to run a new tool day-to-day?
3. Is there interest/willingness among staff and management to adopt an AI-based tool, independent of whether the capacity exists yet?
4. How does Thuiszorg Totaal currently divide its budget between maintaining existing care operations and piloting new tools or ways of working, and what does that ratio suggest about how much financial slack actually exists for an AI investment, independent of whether the tool itself is affordable?
5. How would Thuiszorg Totaal's elderly clients (and their families) actually want to be notified, e.g. via app, SMS, phone call, and what does that imply for adoption risk?

4. Quality criteria for the handbook.

- The handbook mirrors our AEL platform's output qualities
- It is clear to a non-technical owner
- It is accurate against sources we've appraised
- Honesty about responsible AI and trust issues (especially with elderly, health-adjacent client data) is upheld.
- It is useful enough that Thuiszorg Totaal's management could act on it
- A chapter in the handbook is included that ends in concrete first steps rather than general commentary.

5. Methodology.

For the external questions 1 and 2 we will use Desk Research with CBS data on home-care staffing and digitisation, Thuiszorg Totaal's own Chamber of Commerce and annual-account filings, ActiZ (the sector body) and other branch publications, and regional Gelderland press on care-sector digitisation. Vendor material is used only to establish what tools exist, not as evidence that they work.

For the external Q1 specifically, we also search for published case material, sector-press accounts, or referenceable adopters of comparable scheduling tools in home care or similar field-service settings, and we report that search whether or not it turns up a usable case.

Internal Q4 about Thuiszorg Totaal's budget split between operations and new initiatives is addressed primarily through desk research on Thuiszorg Totaal's available financial filings, supplemented by a direct question to management during the site visit if the filings don't give us enough resolution.

Field research consists of one visit to Thuiszorg Totaal in weeks 2–4, serving Internal Q1, Q2, Q3 and Q5. We will interview a planner and, if possible, caretakers, in two distinct parts:

first factual questions about current visit-planning and communication practice (Q1) and the systems and technical capacity already in place (Q2), then separate, explicitly attitudinal questions about interest in and concerns about adopting an AI-based tool (Q3) which will be asked independently of the capacity questions.

We will also interview a small number of Thuiszorg Totaal's elderly clients directly, where consent and access allow, to understand their actual preferred notification channel and what that implies for adoption risk (Q5). If direct client access cannot be arranged in time, we will ask Thuiszorg Totaal staff for their assessment as a fallback, and flag this explicitly as a proxy rather than direct evidence.

Week 1 AIBS proposal - Response to the Socratic gate

The Socratic gate on our week 1 proposal raised five questions, each of which shaped this revision. In response to the concern that STMG (1,200+ staff) falls well outside the SME range this course targets, we have reframed STMG as the source of our initial hypothesis only, and confirmed Thuiszorg Totaal, a home-care organisation of approximately 70 staff, as our planned field-research partner and the actual subject of the handbook throughout. The question of whether a tool's country of origin affects the GDPR obligations it carries is now built into our external analysis as its own sub-question. We have also separated what was previously one bundled internal question into two: whether Thuiszorg Totaal has the technical capacity to run a new tool, and, independently, whether staff and management are actually willing to adopt one. Each is now investigated through its own interview questions. Our desk research methodology commits us to actively searching for a documented case of a comparable provider that has adopted a similar tool, and to reporting transparently if no such case can be found, rather than relying on vendor claims alone. Finally, on the question of whether Thuiszorg Totaal was chosen for genuine SME fit or primarily for field access: it is realistically a combination of both, and we mitigate the resulting risk by cross-checking Thuiszorg Totaal's reported costs and adoption barriers against sector-level data from CBS and ActiZ, rather than treating its account as representative on its own.

